

□ Introduction

- LEO constellation and Space Sustainability
- Post-mission Disposal & Satellite Materials

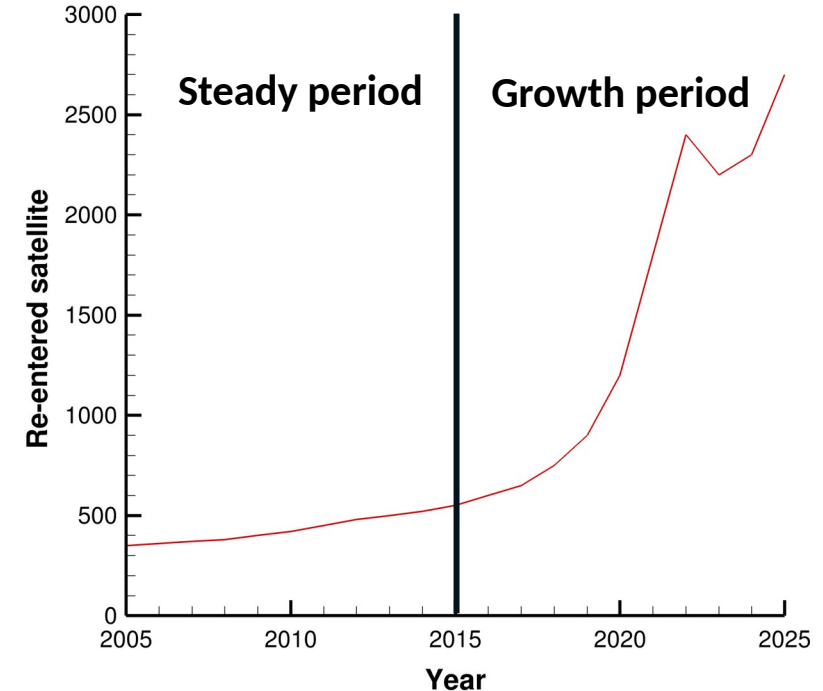
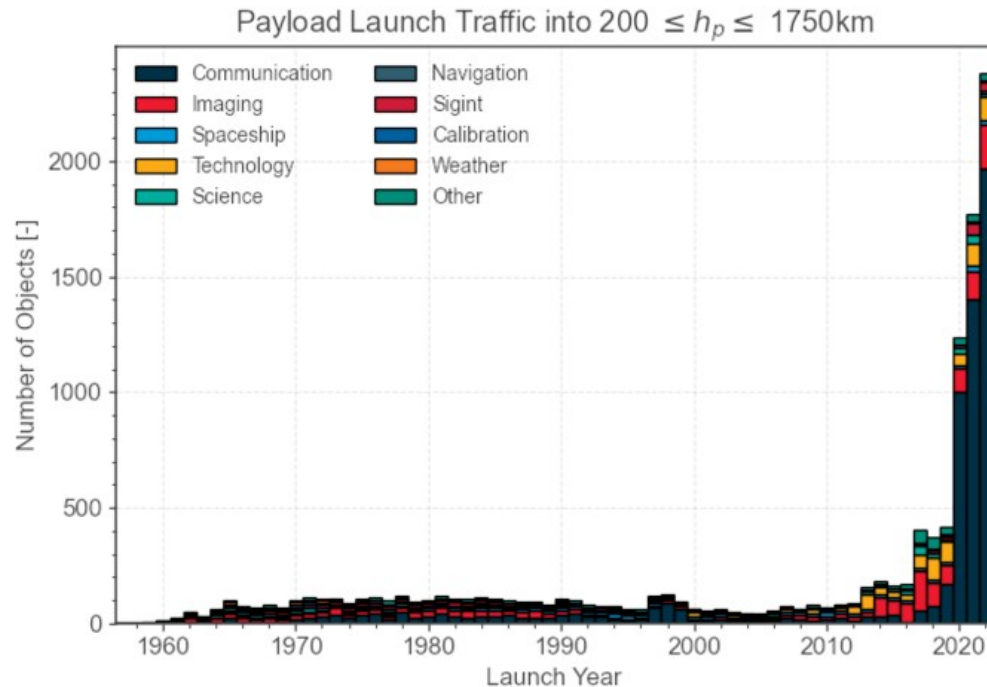
□ Multi-scale Modelling Framework

□ Re-entry Emissions

- Direct and In-direct emissions
- Species and size distribution
- Sensitivity on Re-entry Conditions

□ Summary and Conclusion

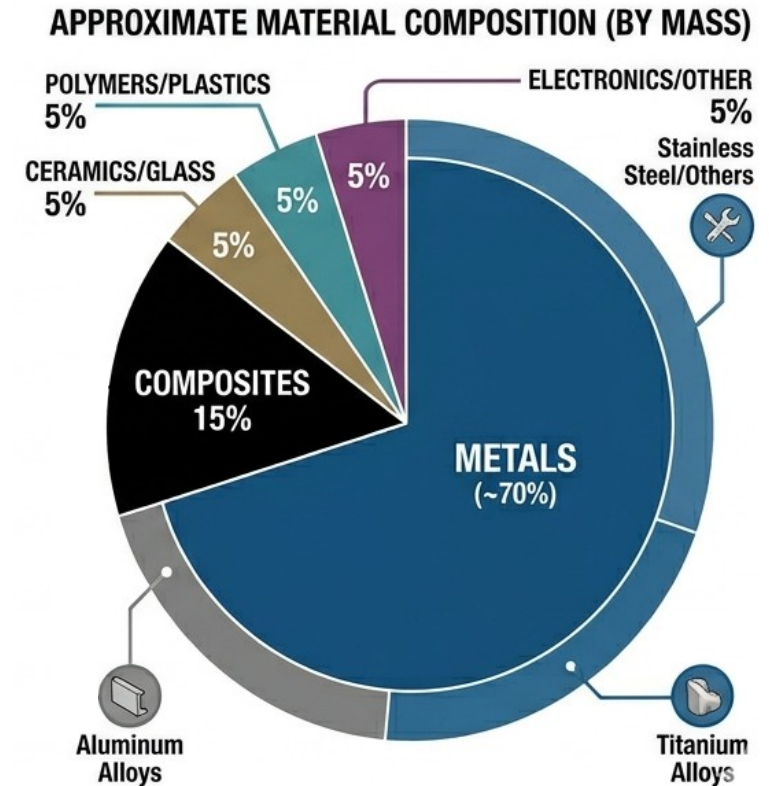
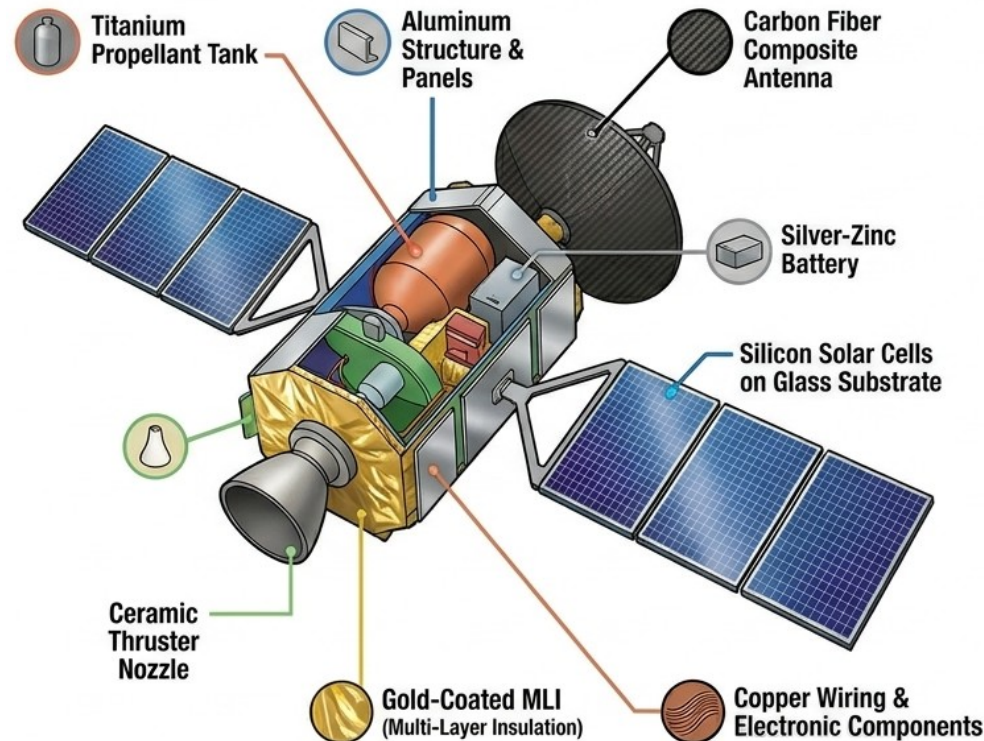
LEO Space Activities and Space Sustainability



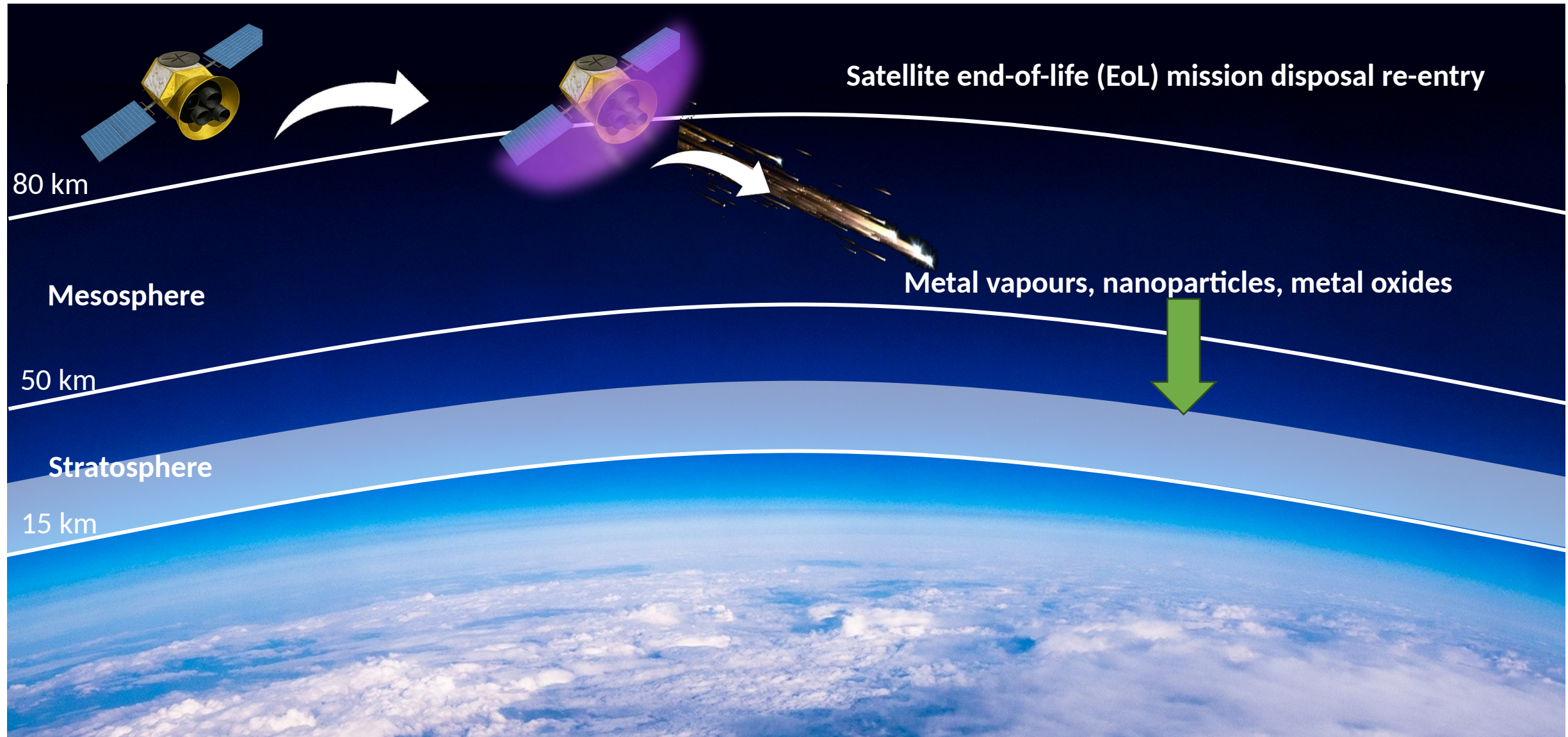
- ❑ Total Year-on-Year (YoY) growth in satellites launched in LEO averaged over the **last 5 years is 39%**.
- ❑ The global launch industry is projected to deploy more than **43,000 new satellites annually** by 2035.
- ❑ Post-mission disposal (PMD) in Low Earth Orbit (LEO) is a key requirement for mitigating space debris and ensuring the long-term sustainability of space operations
- ❑ The deployment of large LEO constellations could result in as many as **10,000 re-entries per year**.

Satellite Materials and Ablation

- ❑ Satellite structures use a diverse set of materials selected to withstand the extreme conditions of space, since no single material can endure the full range of thermal, mechanical, and radiation environments encountered in orbit.
- ❑ Approx 60 ~ 70% of a satellite's mass consists of metals, with aluminium as the dominant material.
- ❑ During re entry, intense aerodynamic heating raises the surface temperature of the satellite, causing materials to melt, decompose, and vaporise (known as 'ablation')
- ❑ Fragmentation increases surface area and accelerates burn up, leading to more complete destruction of smaller components.



Influence of Re-entry Emissions



Atmosphere dynamics and circulation

- ❑ Model the transport of re-entry emission in the middle and upper atmosphere
- ❑ **CESM** (Community Earth System Model) version 2.1.5 with **WACCM** (Whole Atmosphere Community Climate Model)
- ❑ Residence time of re-entry emissions
 - A key metric for assessing the environmental impact of re-entry emissions
 - Defined as the average time a particle or a parcel resides within a given atmospheric layer, from its initial injection to until its removal through atmospheric transport.

Previous Approach

- Use Cunningham-corrected Stokes equation.
- By integrating the viscous drag force into a 1D atmospheric model based on the US Standard Atmosphere, the residence time of re-entry emissions is estimated under the assumption that gravitational settling is the dominant vertical transport mechanisms.

$$\vec{F}_v = -\frac{3\pi\mu d_a}{C_c} \vec{v} \quad C_c = 1 + K_n(1 + 1.00 + 0.570 e^{-1.144/K_n})$$

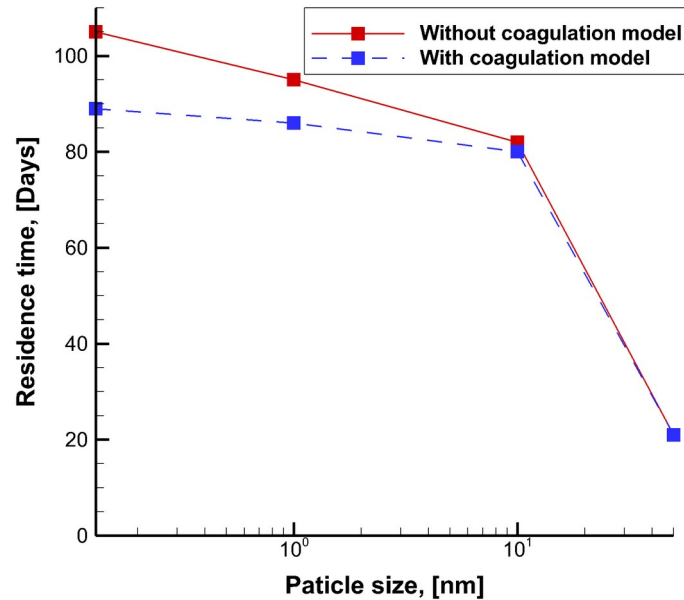
➡ Residence time from 80 km and 50 km : approximately 30 year

✓ **Atmospheric circulation and eddy diffusion** also play important role in determining the transport and distribution of the re-entry particles.

The simulated particle size distribution is incorporated into the Community Aerosol and Radiation Model for Atmospheres (CARMA) as a new aerosol source term and species.

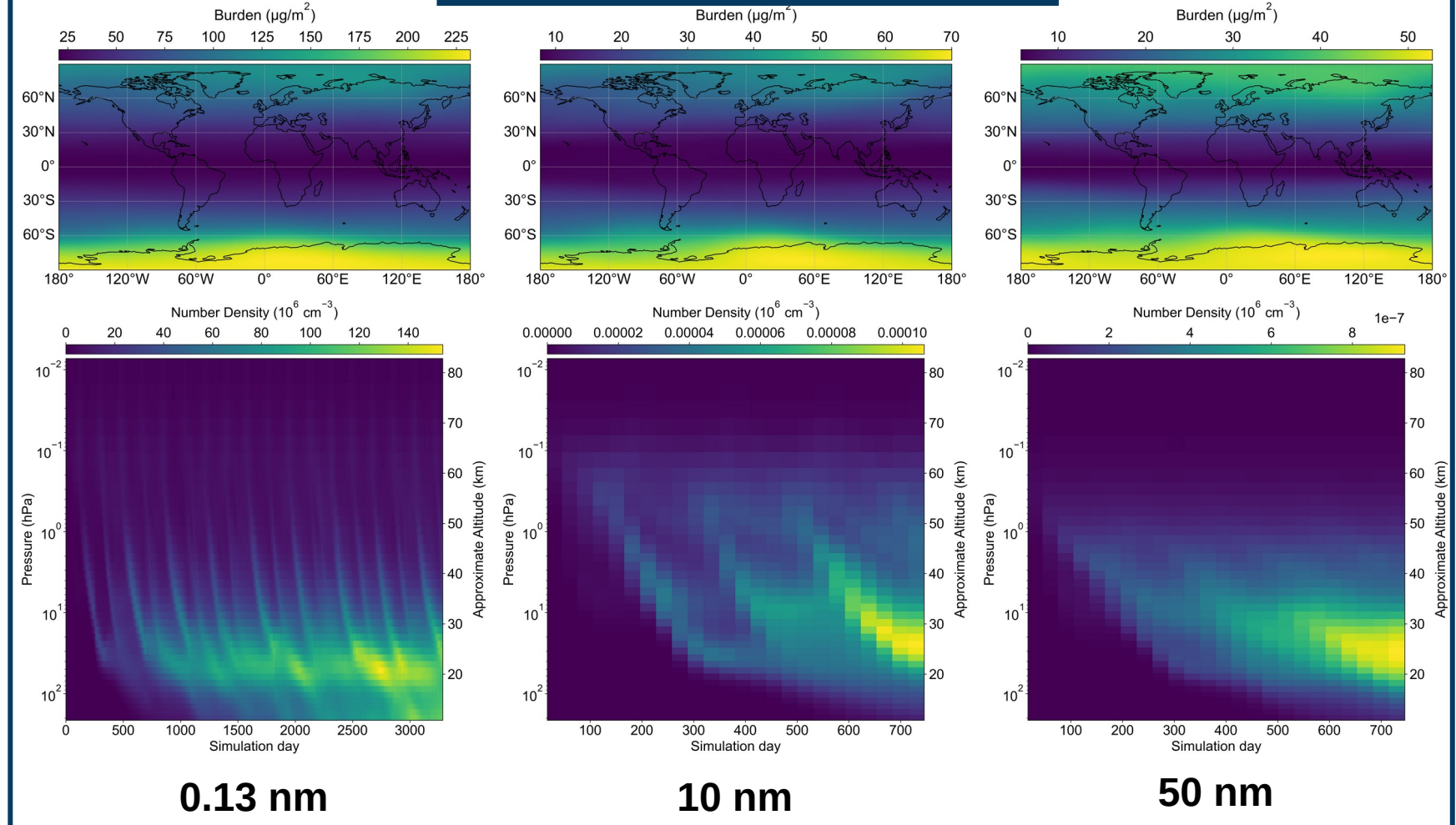
Residence time and Global distribution

□ **Residence time** $\equiv$ the time required for a particle to reach the stratosphere (50 km)

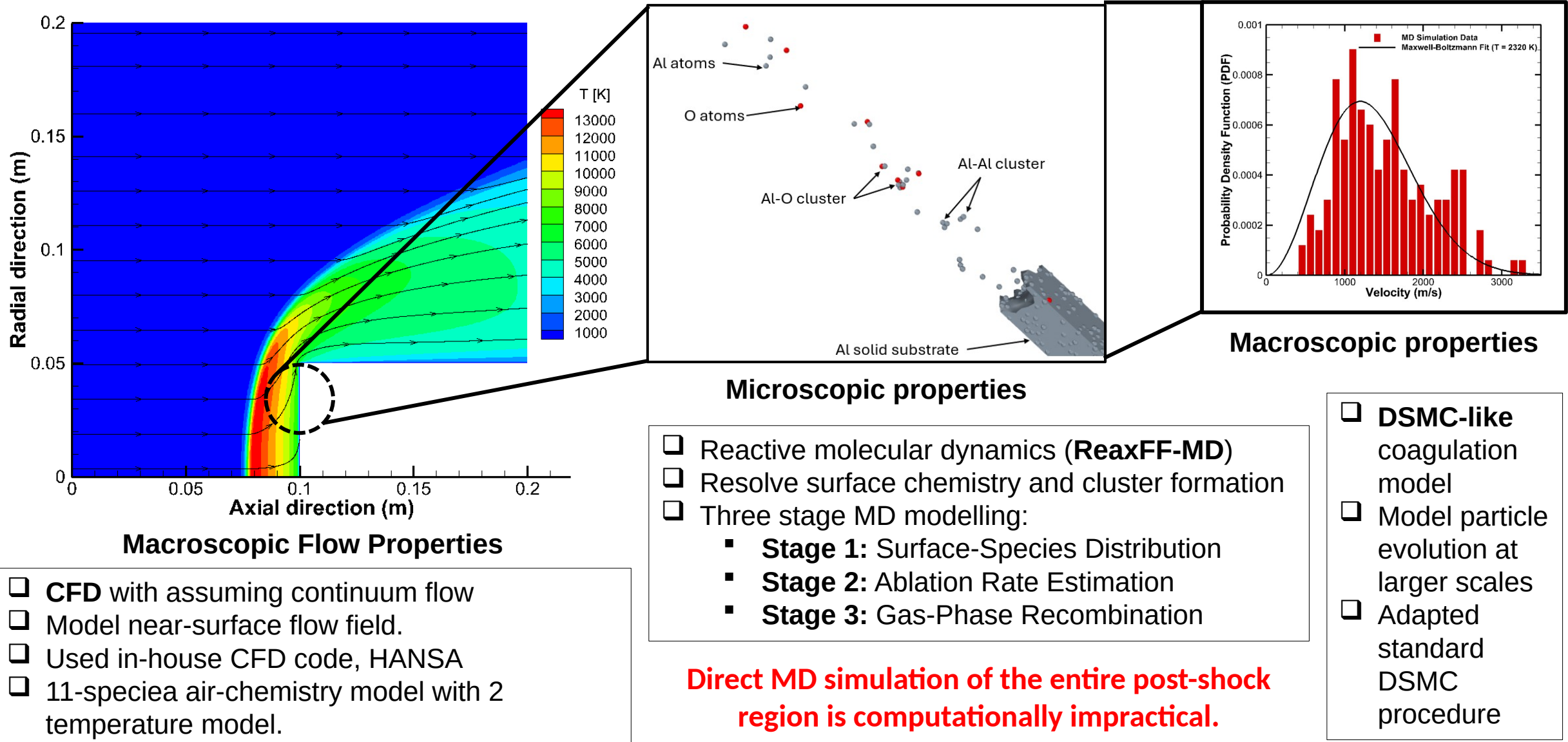


□ The danger lies in the **catalytic nature** of metal oxides.

Without Coagulation Model



Overview of CFD-MD Ablation Modelling



Macroscopic Flow Property Modelling

□ Governing equations

- Mass , momentum and total energy conservation equations
- 11-species air chemistry model with multi-temperature model

$$\frac{\partial \mathbf{Q}}{\partial t} + \nabla \cdot (\mathbf{F} - \mathbf{F}_v) = \mathbf{S}_{NEQ}$$

$$\mathbf{Q} = \begin{pmatrix} \rho_1 \\ \rho_2 \\ \vdots \\ \rho_s \\ \rho u \\ \rho v \\ \rho w \\ E \\ E_{v1} \\ E_{v2} \\ \vdots \\ E_e \end{pmatrix} \quad \mathbf{F} = \begin{pmatrix} \rho_1 u \\ \rho_2 u \\ \vdots \\ \rho_s u \\ \rho u \mathbf{U} + \delta_{ij} p \\ \rho v \mathbf{U} + \delta_{ij} p \\ \rho w \mathbf{U} + \delta_{ij} p \\ (E + p) \mathbf{U} \\ E_{v1} \mathbf{U} \\ E_{v2} \mathbf{U} \\ \vdots \\ (E_e + p_e) \mathbf{U} \end{pmatrix} \quad \mathbf{F}_v = \begin{pmatrix} -J_{i,1} \\ -J_{i,s} \\ \vdots \\ -J_{i,s} \\ \tau_{i,x} \\ \tau_{i,y} \\ \tau_{i,z} \\ \tau_{i,j} \mathbf{U}_j - \left(q_{tr} + \sum_s^{all\ species} q_{v,s} \right) - \sum_s^{all\ species} J_{i,s} h_s \\ -q_{v1} \\ -q_{v2} \\ \vdots \\ q_e \end{pmatrix} \quad \mathbf{S} = \begin{pmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \vdots \\ \dot{\omega}_s \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ S_{v1} \\ S_{v2} \\ \vdots \\ S_e \end{pmatrix}$$

Conserved variable Inviscid flux vector Viscous flux vector Source term

□ Multi-vibrational energy source term for molecular species i is modelled as:

$$\mathbf{S}_{v,i} = \mathbf{S}_{chem,i} + \mathbf{S}_{v-t,i} + \mathbf{S}_{v-v,i}$$

where $\mathbf{S}_{chem,i}$ = vibrational energy gained or removed due to chemical reactions
 $\mathbf{S}_{v-t,i}$ = vibrational-translational energy exchange
 $\mathbf{S}_{v-v,i}$ = vibrational-vibrational energy transfer between unlike molecular species

Reduced Vibrational-Vibrational Collisional Probability Model

□ Vibrational-vibrational energy relaxation model

$$S_{v-v,i} = \sum_{j \neq i}^{All\ molecules} P_{i,j} Z_{i,j} m_i (e_{v,i}(T'_{v,i}) - e_{v,i}(T_{v,i}))$$

□ Traditional multi-vibrational temperature model used the constant V-V collisional probability as 0.01.

- Underestimating relaxation time below 13000 K and Overestimating at high temperature

□ Accuracy can be improved using full Schwartz-Slawsky-Herzfeld (SSH) theory, but computational expensive

➡ **Need a better V-V collisional probability model with low computational cost**

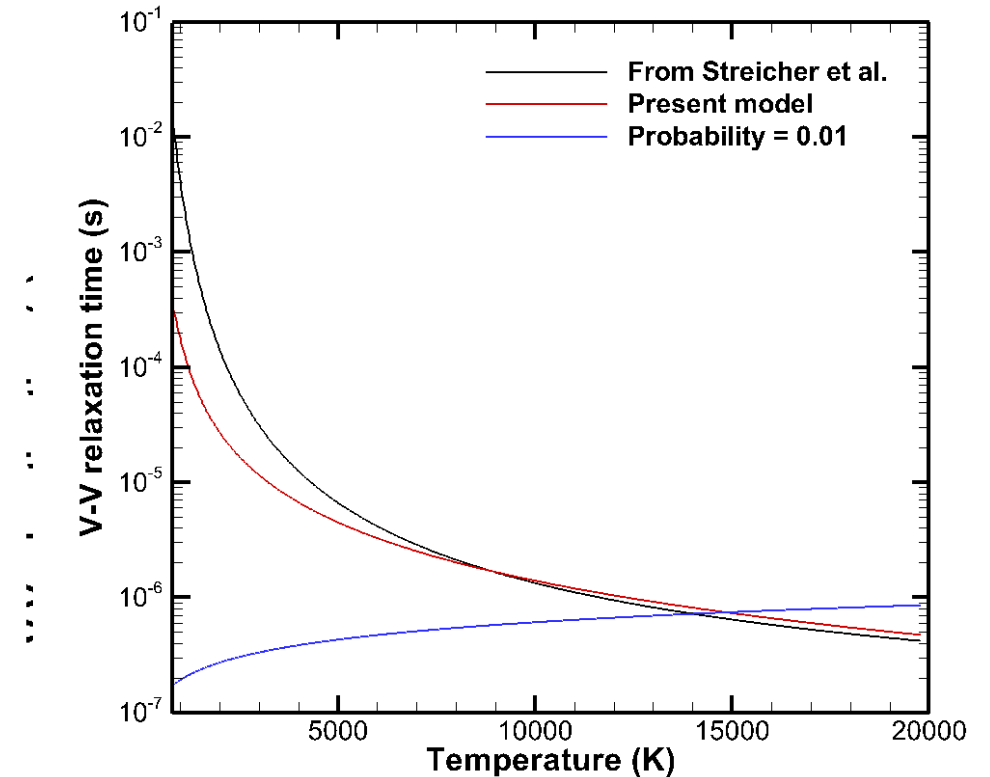
□ Based on on Schwartz-Slawsky-Herzfeld (SSH) theory, we propose a simplified V-V collisional probability model for CFD as $AT^n + B$

- $P_{N_2-O_2} = 1.622 \times 10^{-14} T^{2.96} + 4.015 \times 10^{-10}$
- $P_{N_2-NO} = 1.6936 \times 10^{-8} T^{1.38}$
- $P_{O_2-NO} = 1.6936 \times 10^{-8} T^{1.38}$

$P_{i,j}$ = Collision Probability for V-V energy transfer

$Z_{i,j}$ = Collision number

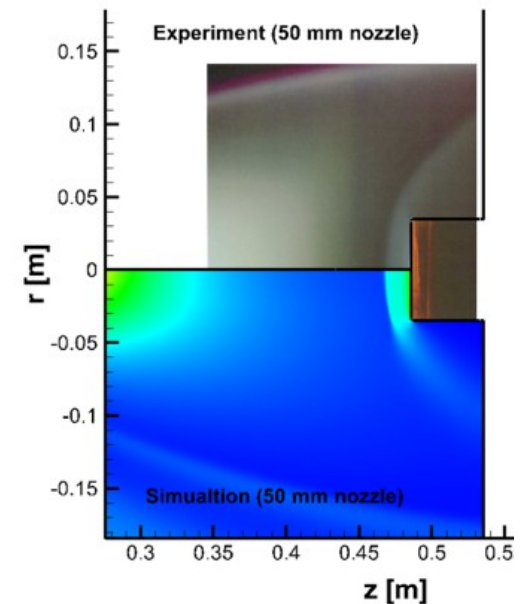
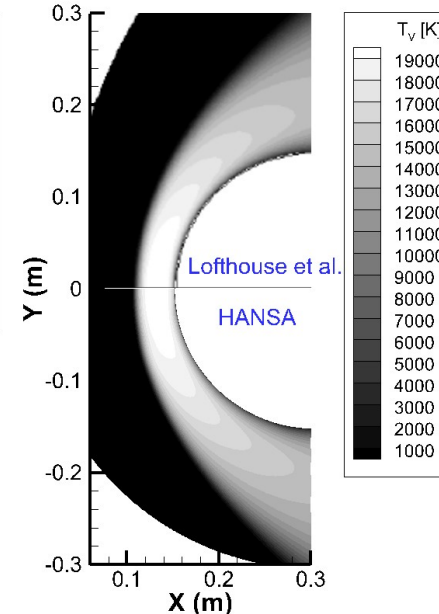
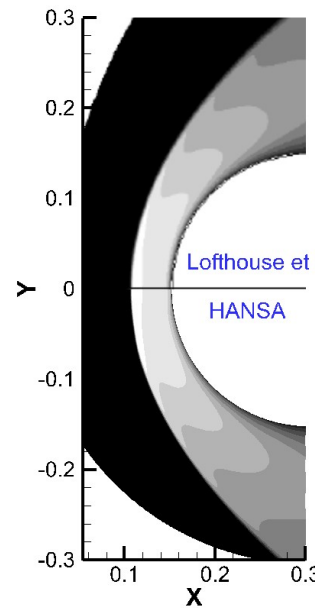
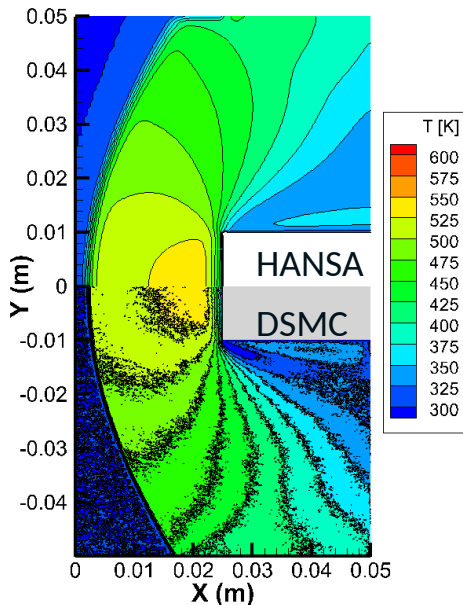
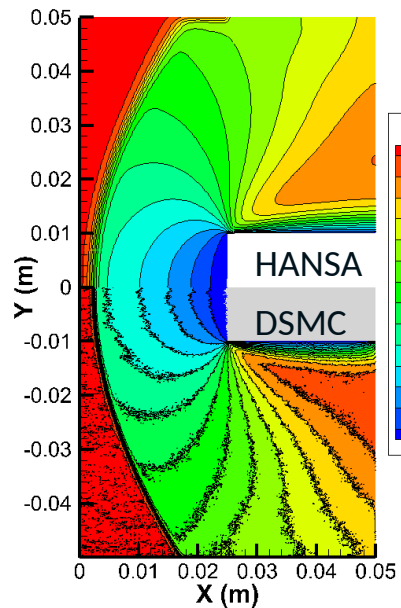
$T'_{v,i}$ = Vibrational temperature of the two collision partners



V-V relaxation time for N₂-O₂ system at 1 atm pressure

In-house CFD Solver: HANSA

- ❑ HANSA (Hypersonic Aerothermal Non-equilibrium flow Solver for an Arc-heated wind tunnel) is an in-house thermo-chemical non-equilibrium CFD code developed by the University of Southampton
 - ⊗ A general 2D/3D hybrid MPI/OpenMP unstructured CFD solver using a finite volume method.
 - ⊗ Originally developed to model a plasma wind tunnel and later expanded to general hypersonic flows.
 - ⊗ Recently adapted to an unstructured Cartesian mesh with AMR.
 - ⊗ Material response model is available.



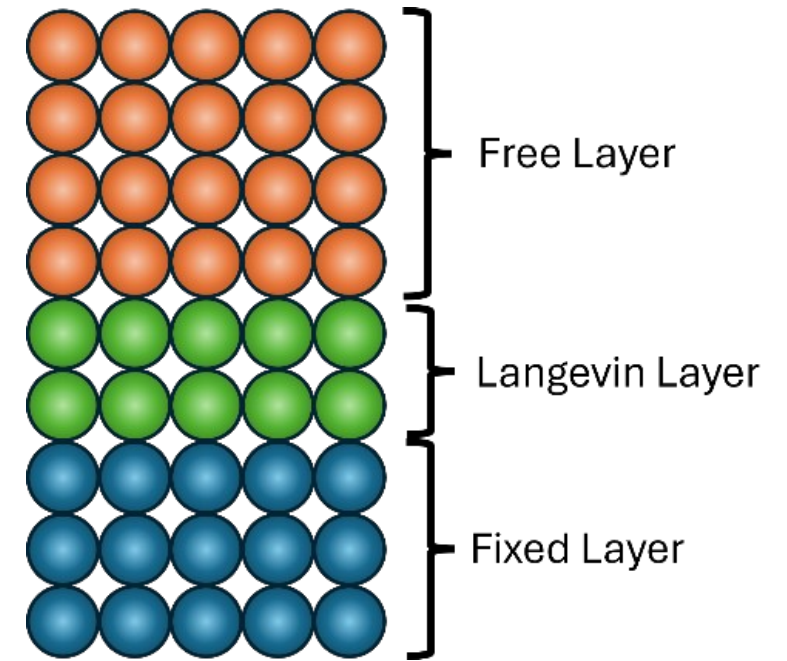
Modelling of Multi-Temperature Vibrational Model for
Hypersonic Flows

Microscopic properties: Molecular Dynamics (MD)

- Three distinct stages of molecular dynamics (MD) simulations are performed.

Stage 1

- Determine the species distribution **at the surface**,
- Consist of three functional layer (Free / Langevin / Fixed)
- Free layer (NVE)**: Enabling natural evolution of the surface under oxygen bombardment without any external controls
- Langevin layer (NVT)**: Controls the substrate temperature at fixed volume during heating and cooling.
- Fixe layer (NPT)**: allows structural relaxation by adjusting the simulation cell to account for thermal expansion.
- Aluminium substrate + Impinging Oxygen atom
- Domain: $24.77\text{\AA} \times 24.77\text{\AA} \times 800\text{\AA}$



Microscopic properties: Molecular Dynamics (MD)

- ❑ Three distinct stages of molecular dynamics (MD) simulations are performed.

Stage 2

- Determine ablation rate
- Treated as a **single region** without fixed or thermostatted layer
- Allows the surface temperature to evolve naturally based on internal energy balance and mass loss.
- Statistical time-skipping approach has been used.
- Al ablation flux, ϕ_{Al} , is calculated from

$$\phi_{Al} = \frac{\frac{d(N_{Al})}{dt}}{\frac{d(N_O)}{dt}} \phi_O$$

where

ϕ_O = an incident atomic oxygen flux obtained from CFD results

$\frac{d(N_{Al})}{dt}$ = an atomic Al emission rate

$\frac{d(N_O)}{dt}$ = the incident rate used in the MD through time-skipping approach

Stage 3

- Simulate the gas-phase reaction between an ablated aluminium atom and a dissociated oxygen atom in the gas phase.
- Gas phase recombination (ablated Al atoms and dissociated O atoms under post shock hypersonic conditions)

Microscopic Statistical Model (DSMC-like Coagulation Model)

❑ Models the coagulation of Al-O vapour into clusters (particles)

- Transferring microscopic properties (MD simulation results) to macroscopic properties (re-entry emissions)
- MD (Microscopic properties): capture surface chemistry, bond breaking, and early-time emission statistics.
- Aluminium ablation produces a dense vapour plume consisting of atomic aluminium (Al) and aluminium monoxide (AlO) clusters.
- Statistically sampling collisions and sticking events → Particle number decay and final particle size distributions.

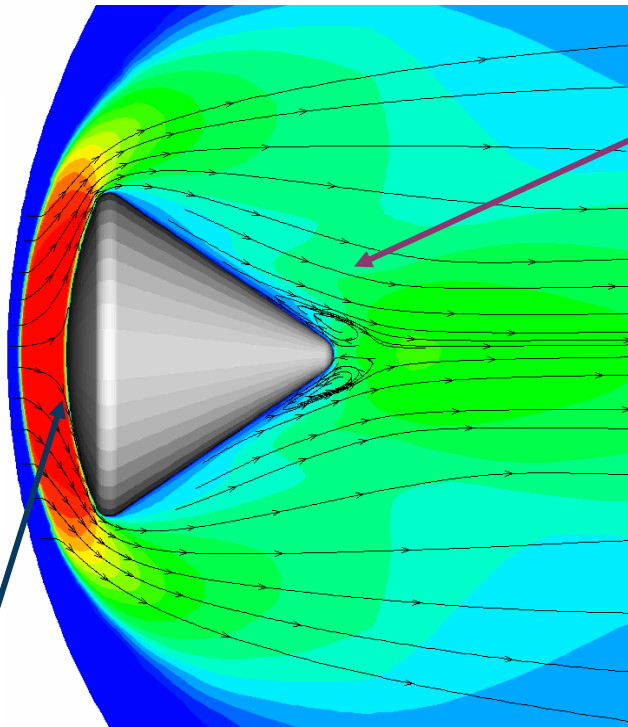
❑ Assumption

- **Particle motion** and **collisions** are decoupled over a time interval shorter than the mean collision time.
- Particles are initially distributed randomly with velocities sampled from a **Maxwellian distribution**.
- All particles undergo ballistic motion without external forces

❑ Procedures

- Using Bird's no-time counter (NTC) method, calculate the total number of possible collisions: $N_{coll} = \frac{N(N-1) (\sigma c_r)_{max} \Delta t}{2 V}$
- Collision pairs are selected randomly and accepted using an acceptance-rejection methodology: $P_{coll} = \frac{\sigma c_r}{(\sigma c_r)_{max}}$
- Once the system reaches a statistically steady state, ensemble-averaged statistics are collected over an extended sampling period
- Upon selecting a collision pair, a criterion determines if coagulation occurs based on an energy-dependent sticking probability.
- Coagulation is accepted using an acceptance-rejection scheme with probability.

Direct and In-direct emissions

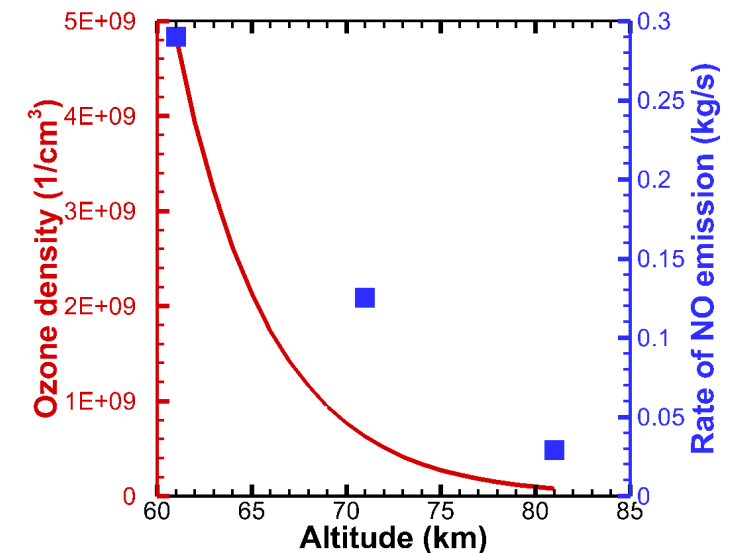
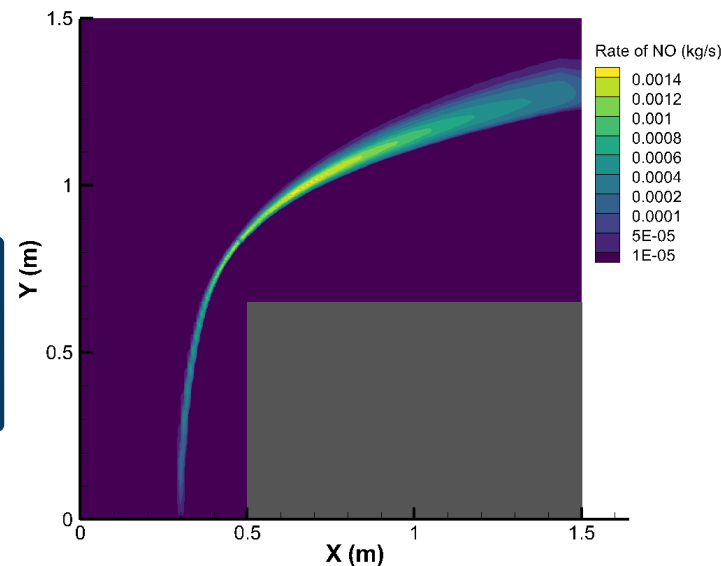


Direct emission

- ❑ Ablated satellite materials and their by-products

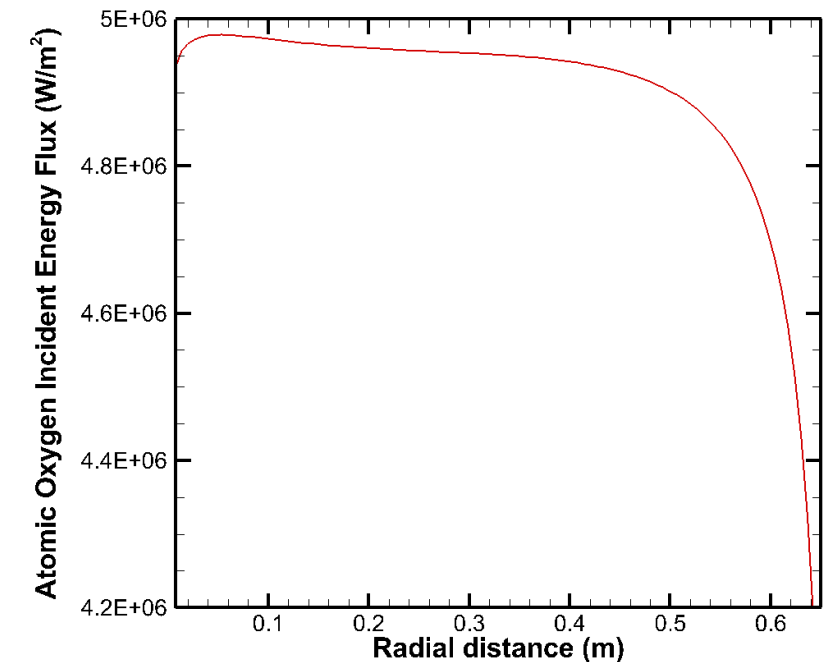
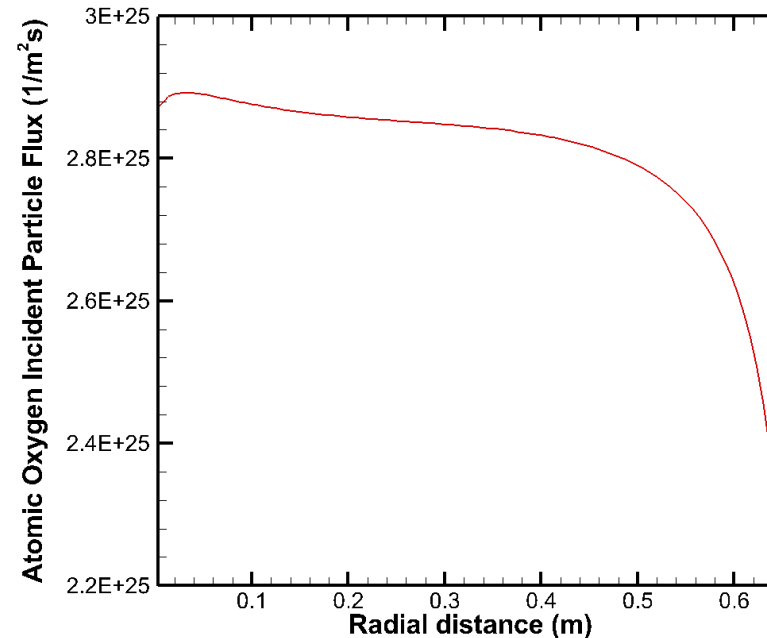
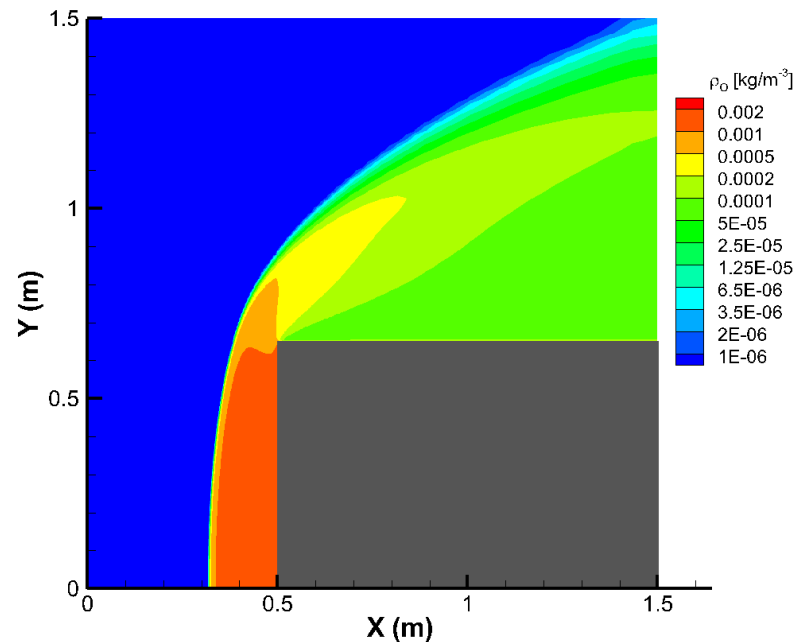
In-direct emission

- ❑ Nitric oxide by the dissociation of air
 - The net NO production is obtained by integrating the volumetric NO production rate from the flow-field results.
 - Highly localised near the bow shock, reflecting the dominance of Zeldovich reactions under high temperature condition.
 - Ultraviolet photolysis efficiently removes NO in sunlight, but this process is suppressed during polar winter.



Flow Properties

- ❑ Atomic oxygen governs the chemical response of the satellite surface.
- ❑ Flow properties including gas temperature, atomic oxygen number density, and bulk velocity are extracted from the CFD solution at locations immediately adjacent to the wall.
- ❑ Gas-surface interactions are assumed to occur in the free molecular regime.
 - The mean free path of atomic oxygen in the near-wall region is comparable to the characteristic surface roughness scale

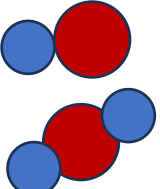


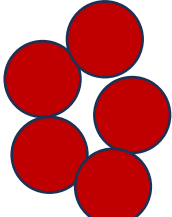
Microscopic Properties from MD

□ Stage 1 (Surface Chemistry)

- Provide the fractions of emitted pure Al atoms, Al-Al clusters, and Al-O clusters as functions of this ratio.
- At each time step, a thin near-surface layer of aluminium is assumed to participate in reactions.

 Aluminium atoms → Aluminium vapour

 Aluminium & Oxygen clusters → Aluminium oxides

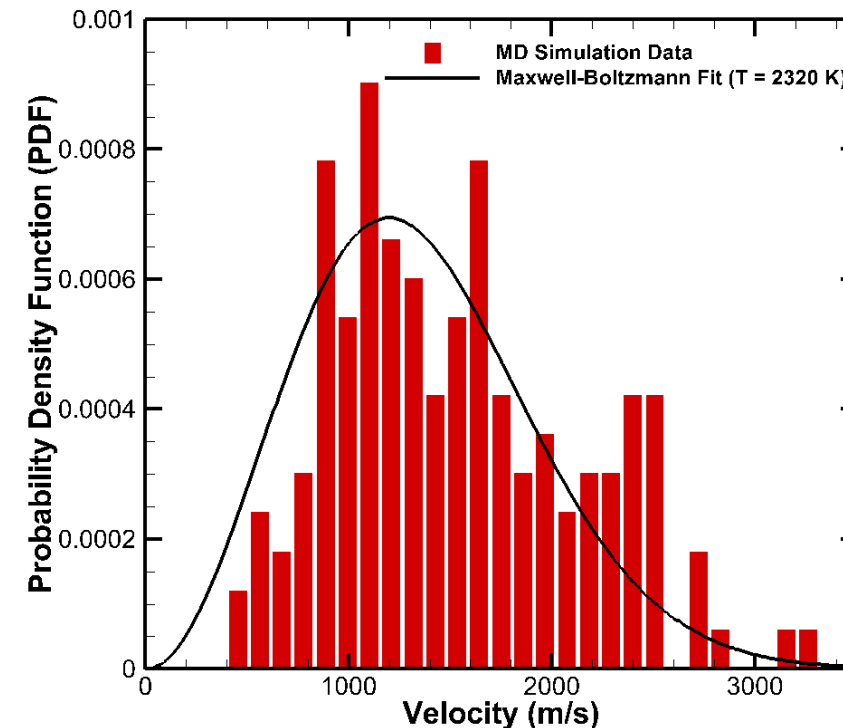
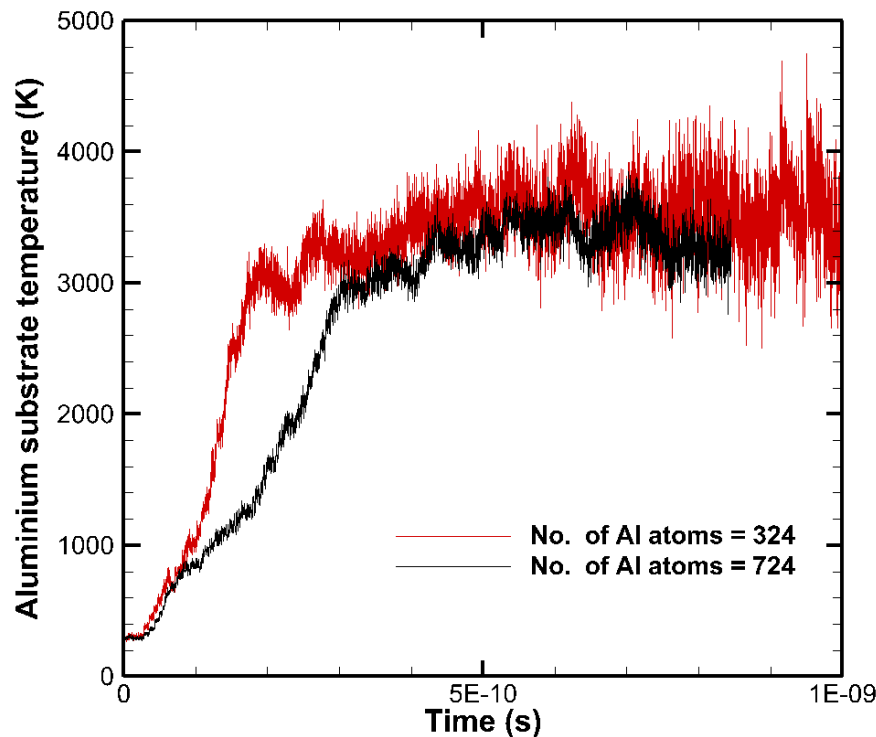
 Aluminium clusters → Aluminium nanoparticle

- For 274 kg aluminum satellite, the results yield **212.1 kg of pure Al atoms**, **68.67 kg of Al-Al clusters**, **46.23 kg of Al-O clusters**, and 27 kg of oxygen atoms.

Microscopic Properties from MD

□ Stage 2 (Ablation Rate Estimation)

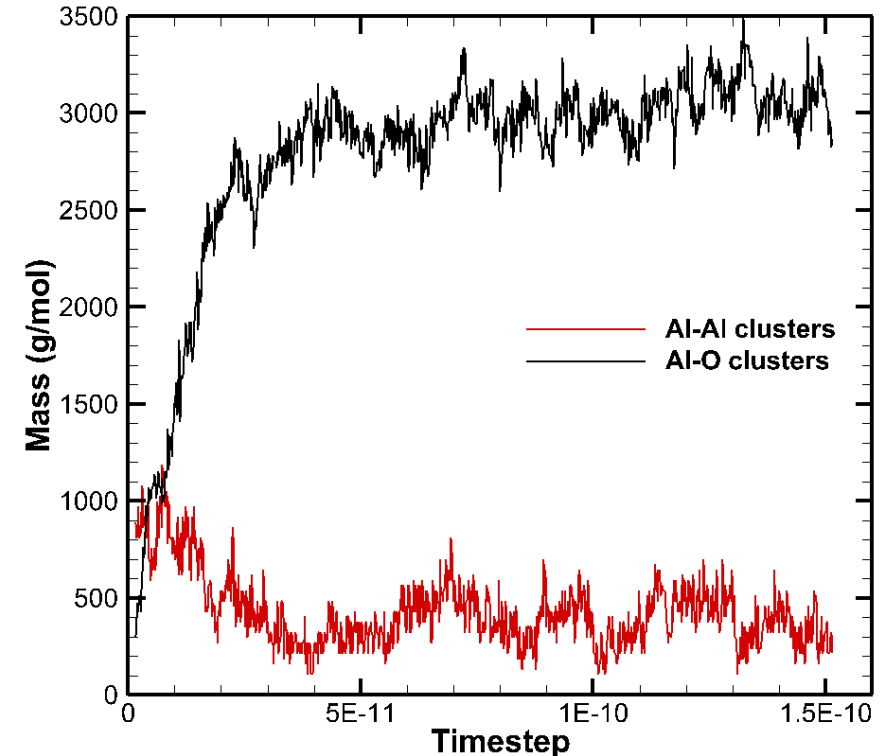
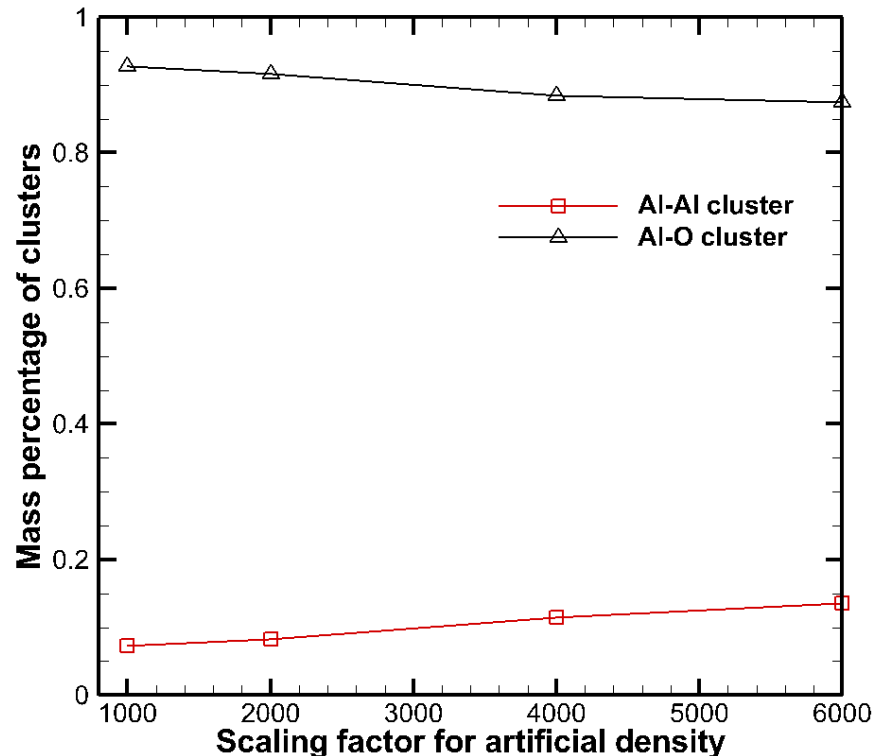
- Quantify the aluminium ablation rate resulting from atomic oxygen energy flux.
- The entire aluminium slab is treated uniformly without fixed or thermostatted regions.
- For evaluating the sensitivity of size, utilise two different substrate sizes, consisting of 324 and 724 Al atoms



Microscopic Properties from MD

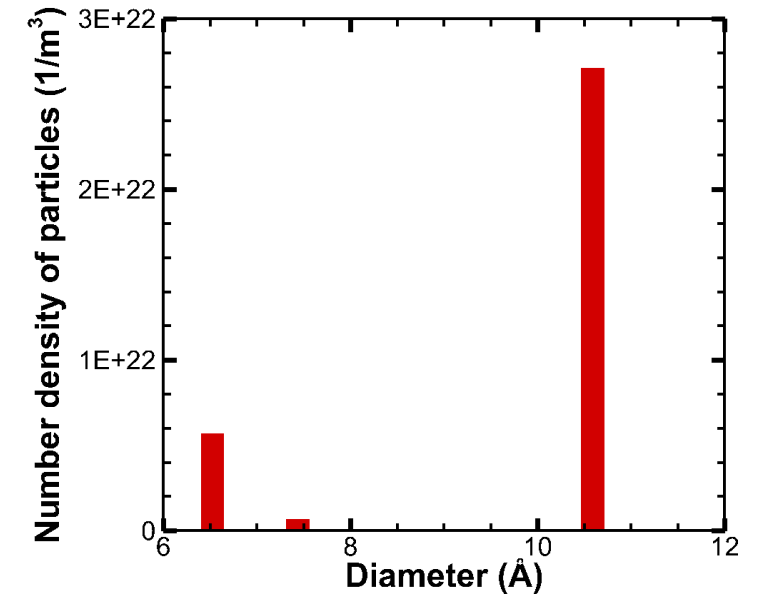
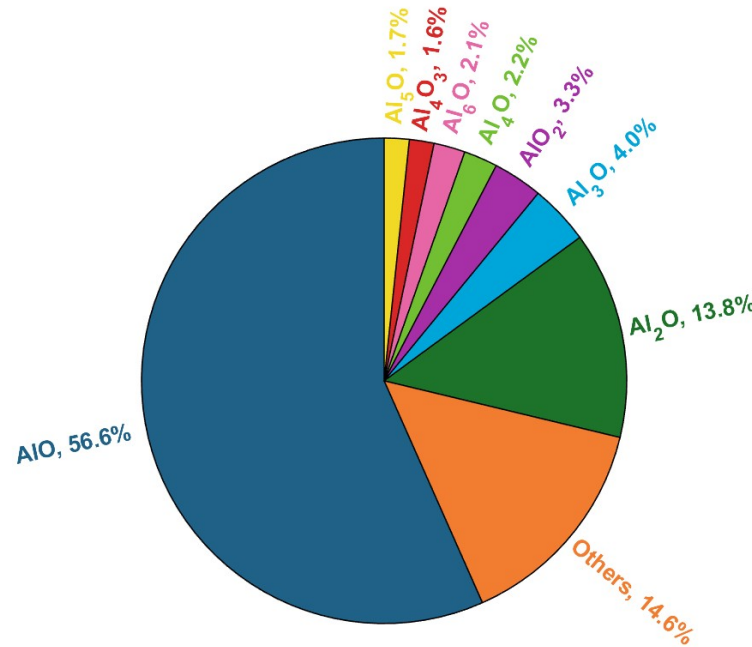
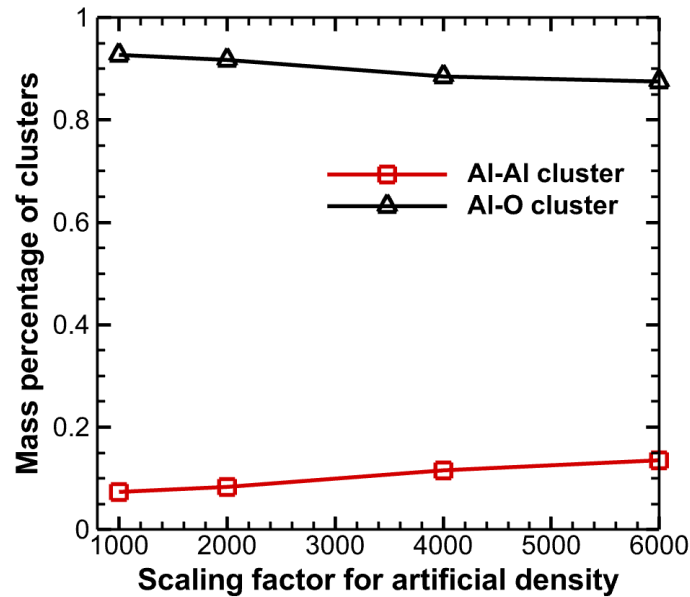
■ Stage 3 (Gas-phase Al-O Reaction)

- Models gas-phase reactions between ablated aluminium (Al) atoms and atomic oxygen (O) in the shock region.
- Steady-state Al and O number densities are derived from the Stage 2 ablation flux
- Simulates a representative volume of the shock-layer region.
- Time-averaged quantities determine the fractions of Al remaining as free atoms and those forming Al-O clusters



Estimation of Total Cluster Formation and Gas-Phase Coagulation

- Combining the surface cluster formation data from Stage 1 with the gas-phase reaction statistics from Stage 3
- Estimate the total Al-O cluster production over the satellite's re-entry.
- Using coagulation model, estimated the physical growth of metal oxide molecules
- Assumption:** complete burn-up of the aluminium outer casing.



- The total mass of Al-O clusters produced during re-entry is approximately **74 kg** (Satellite mass: 274kg)

Sensitivity on Re-entry Conditions

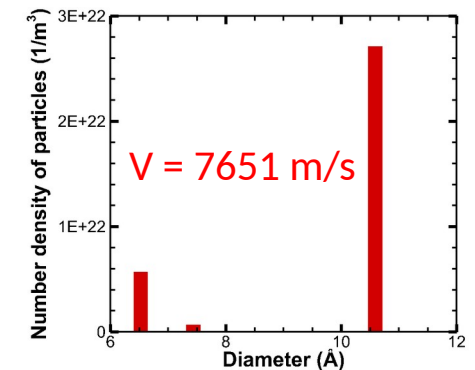
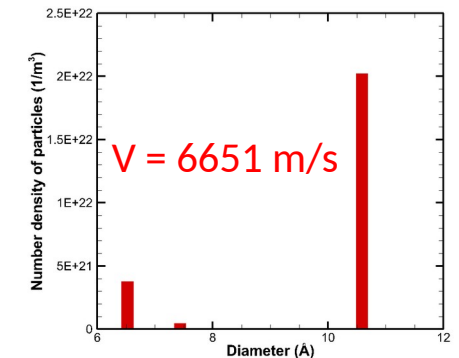
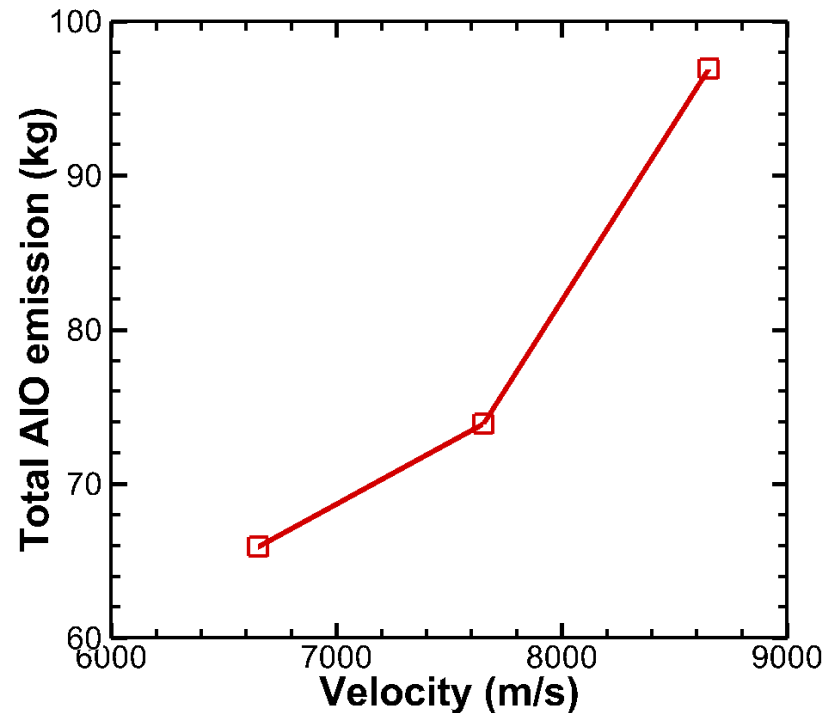
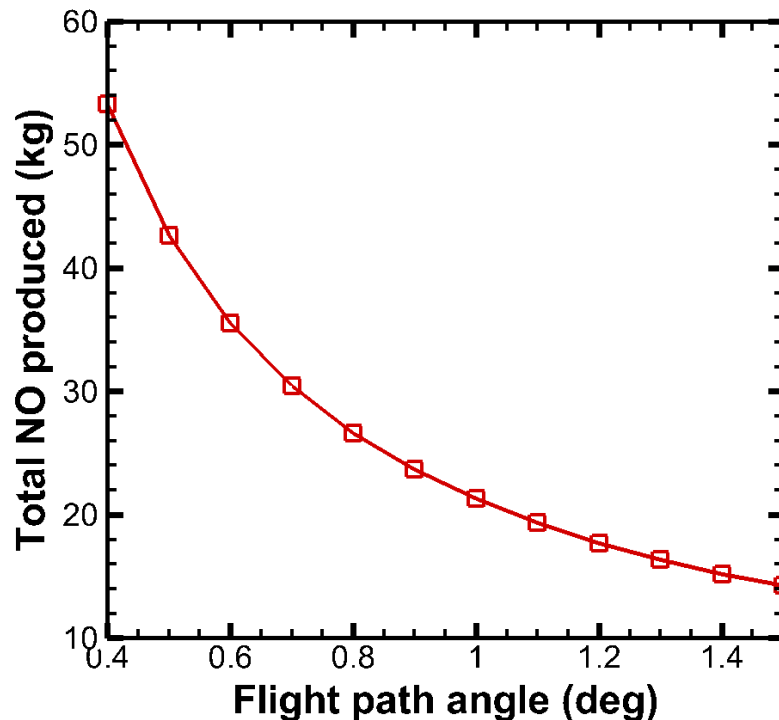
- The average heating rate ($\dot{q}$) and the altitude (h_{peak}) at which peak heating occurs during atmospheric ballistic re-entry can be approximated as:

$$\dot{q} = k \left(2B\alpha \sin \gamma_0 \ln \frac{V}{V_0} \right)^{0.5} V^3$$

$$h_{peak} = \frac{1}{\alpha} \ln \left(\frac{\rho_s}{B\alpha \sin \gamma_0} \right)$$

where

$\frac{1}{\alpha}$ = scale height of an atmospheric model
 ρ_s = surface density
 γ_0 = re-entry angle
 B = Ballistic coefficient
 k = constant related to the geometry of vehicle



Key Findings

- ❑ The rapid growth of LEO constellations could increase atmospheric re-entry rates to more than 10,000 satellites per year over the coming decades.
- ❑ Multi-scale modelling predicts that satellite re-entry generates a diverse range of emissions, including metal vapours, metal oxide molecules, clusters and nanoparticles, with aluminium oxide identified as a dominant product.
- ❑ Nano-scale particles (<10 nm) are transported primarily by atmospheric circulation and can remain in the middle atmosphere for extended periods.
- ❑ Larger particles (>50 nm) are increasingly governed by gravitational settling, leading to more rapid transport into the stratosphere.
- ❑ Atmospheric transport and particle evolution are strongly dependent on the initial particle size distribution, coagulation processes and re-entry conditions.

Implications

- ❑ Satellite re-entry introduces a new anthropogenic source of metal species into the middle atmosphere that is expected to increase substantially with future constellation deployment.
- ❑ The long residence time and global transport of nano-scale metal oxides indicate that their environmental impact cannot be assessed using local re-entry analyses alone.
- ❑ Significant uncertainties remain regarding emission inventories, particle evolution, heterogeneous chemistry and long-term atmospheric impacts, particularly on ozone chemistry.

- ❑ This work demonstrates the first integrated multi-scale framework linking satellite ablation physics, nanoscale particle formation and global atmospheric transport, providing a foundation for future environmental impact assessments of large satellite constellations.

Future Work

- ❑ Improve the fidelity of emission source terms through coupled ablation, chemistry and particle evolution modelling.
- ❑ Validate model predictions using laboratory experiments, atmospheric observations and coordinated international measurements.
- ❑ Develop a robust scientific evidence base to support future environmental assessments and international standards for sustainable satellite disposal.

This is a rapidly emerging research area requiring expertise across atmospheric science, aerospace engineering, chemistry and Earth system modelling. We welcome opportunities for international collaboration.

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